A SYMMETRIC ALGEBRA IS SMOOTH EXACTLY WHEN THE MODULE IS FINITE PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Smoothness of a ring map tested through the naive cotangent complex, and the fact that a smooth algebra has finite projective conormal module. the author cites Algebra, Lemma algebra-lemma-section-smooth and Algebra, Definition algebra-definition-smooth at proof lines 2122-2124 and 2134-2136. Supplies the criterion in both directions.
- The cotangent complex of a symmetric algebra presented by a free module. the statement counted by the sibling card S1-algebra-cotangent-complex-symmetric-algebra at 2066-2109, cited at proof lines 2130-2132. Identifies the complex whose concentration in degree zero gives smoothness.
- A finite projective module is a direct summand of a finite free module, so a surjection onto it splits. used at proof lines 2125-2130. Makes the symmetric algebra finitely presented over the base.

Source reference. Original lemma 07M6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9716–9716. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05D5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6943–6943. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9460–9460. Navigation pointer to a statement the

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Source reference. Original lemma 00T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1060–1060. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring. Let M be an A-module. Then $C = Sym^*_A(M)$ is smooth over A if and only if M is a finite projective A-module.*

Proof. Let $\sigma : C \to A$ be the projection onto the degree 0 part of C. Then $J = \mathrm{Ker}(\sigma)$ is the part of degree > 0 and we see that $J/J^2 = M$ as an A-module. Hence if $A \to C$ is smooth then M is a finite projective A-module by Algebra, Lemma 05D5.

Conversely, assume that M is finite projective and choose a surjection $A^{\oplus n} \to M$ with kernel K. Of course the sequence $0 \to K \to A^{\oplus n} \to M \to 0$ is split as M is projective. In particular we see that K is a finite A-module and hence C is of finite presentation over A as C is a quotient of $A[x_1, \ldots, x_n]$ by the ideal generated by $K \subset \bigoplus Ax_i$. The computation of Lemma 07EV shows that $NL_{C/A}$ is homotopy equivalent to $(K \to M) \otimes_A C$. Hence $NL_{C/A}$ is quasi-isomorphic to $C \otimes_A M$ placed in degree 0 which means that C is smooth over A by Algebra, Definition 00T2. $\square$